

SDS Final Report Guidelines

Nurul Lubis

Updated 21.7.2026 — Due 3.8.2026

Over the semester, you will have completed six assignments covering key components of a modular spoken dialogue system. For the final report, combine your work into a single **cohesive** report resembling a scientific paper. The report should go beyond general descriptions and instead **focus on your own experimental setup, insights, and results**.

1 Content & Structure

Your report should be written in scientific style (using LaTeX is strongly encouraged) and include:

1. Abstract – Summarize your overall system and key findings.
2. Introduction – Introduce word embeddings, task-oriented spoken dialogue system and its modules.
3. Experimental Set up – Introduce Simple MultiWOZ21 and **ConvLab3**. Introduce your general experimental set up (hardware, BERT version, etc.)
4. Result and analysis – One section per component. Briefly introduce the model. Summarize the results from your exercise reports and **explain which choices had the biggest impact on system performance**. Is it the embedding, the architecture, some applied heuristic, a hyperparameter, or something else? This is the most important part of the report and will strongly influence your grade.
 - (a) NLU with BERTNLU
 - (b) State Tracking with TripPy
 - (c) Policy with PPO
5. Agentic task oriented dialogue - Briefly describe structured prompting to utilize instruction-tuned LLMs for task-oriented dialogue. Analyze its effectiveness and shortcomings.
6. Conclusion – Reflect on what worked well, challenges, and possible improvements.
7. Appendix - include your example dialogues and LLM prompt here. The examples do not count towards the word limit.

Small adjustments to the structure are welcome if that helps you write a better report. Make sure your report is clear, scientific, and self-contained.

2 Guidelines

- You do not need to run additional experiments. Emphasize **your own** experimental results and analyses from exercises 1-6.
- Focus on specific choices, observations, and outcomes.
- Include references to the specific methods or definitions used in the report.
- Include figures/tables where helpful.
- Max 2000 words (excluding **abstract**, references, figures, equations, and appendix).
- **Include the paths to your Git Repos**
 - Required: private fork(s) of convlab3 and trippy on git.hhu.de with user ‘lubis’ as member (access higher than Reporter), with exercises 2-5 committed **prior** to the final report deadline.
- You may use AI writing assistants to improve your English. Note that the analysis and findings presented must be your own.

3 Submission

Upload the final report with name [LASTNAME]_final.pdf to ILIAS. Provide the word count in ILIAS.